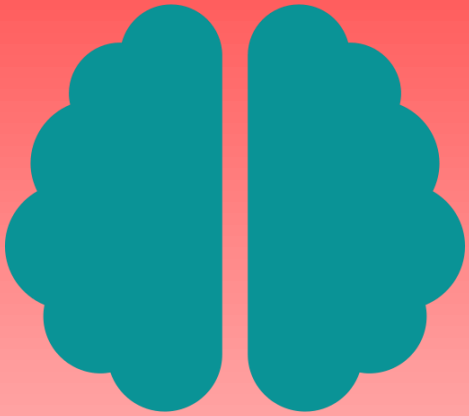




Introduction to LLM Prompting Techniques

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What is a Prompt?

"A prompt is the text (or multimodal input) you send to an LLM to obtain a response. Its structure determines output quality."

Input

What you type — your question, instruction, or context

Processing

The LLM interprets your prompt and generates tokens

Output

The model's response — shaped entirely by the prompt

Why Prompting Matters



Nov 2022

ChatGPT launched — changed everything



Principal

Prompting is the main way users access LLMs



Quality ↑

Better prompts = more accurate, reliable outputs

Tutorial Structure



Part 1

Concepts & Techniques

- ~40–45 minutes
- RTCF Framework
- Few-shot Prompting
- Constraints
- CoT & RAG



Part 2

Hands-on Practice

- Remaining time
- Google Colab notebooks
- Run and debug codes, libraries
- Embedding and Vectors
- Use provided or own data

Techniques We'll Cover



RTCF Framework

Role · Task · Context · Format



Few-shot Prompting

Examples guide the model



Constraints

Control output format



Chain-of-Thought

Step-by-step reasoning



RAG

Ground answers in your data

Before We Start — Setup

You need:

- Gmail account (for Colab access)
- OpenRouter API key (apply before session)
- Basic Python knowledge
- Colab / Jupyter notebook basics

⚠ Data Privacy

- We provide our dataset — open source, safe to use.
- If using your own data: no personal or private information.
- Data is uploaded to Google Cloud and processed via external APIs.
- For sensitive data: run an LLM locally instead.



What You'll Take Away

1

Knowledge

Understand the structure and rationale of each prompting technique

2

Hands-on experience

Apply techniques to real tasks: extraction & data analysis

3

Transferable skills

Prompting skills you can use across research and practical work

4

API access

Know how to connect to OpenAI-compatible APIs via Python

Let's get started!

Why Prompting, Not Fine-tuning?

	Prompting	Fine-tuning
Cost	Free / near-free	GPU compute required
Speed	Instant	Hours to days
Data needed	None (or few examples)	Labelled dataset
Flexibility	Change prompt anytime	Re-train to change
Best for	Most everyday tasks	Domain-specific, large-scale

Hallucination & Bias in LLMs

Hallucination

LLMs can generate plausible-sounding but factually incorrect information. Good prompting (especially RAG and CoT) reduces but does not eliminate this.

Bias

Models inherit biases from training data. Outputs can reflect stereotypes or skewed perspectives. Be critical of results, especially on sensitive topics.

Mitigation via Prompting

Clear instructions, examples, grounding (RAG), and step-by-step reasoning (CoT) all help produce more accurate, balanced outputs.